

MUHAMMAD IZZUN MUSTAQIM BIN ISMAHDI

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 9+ years of experience designing, developing, and deploying full-stack applications across AI, web, and enterprise systems. Proven expertise in Next.js, TypeScript, Python, and Java, with strong track record building AI-powered solutions using OpenAI and Gemini APIs. Skilled in cloud and self-hosted deployment (Docker, VPS, Vercel), database design (PostgreSQL, MySQL, Supabase), and Agile delivery. Experienced in collaborating with global stakeholders, including Japan-based delivery teams, to deliver high-quality, scalable software solutions.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, PHP, C#, C++, SQL, HTML, CSS

Frameworks & Libraries: Next.js, React, Vue, Angular, Django, Flask, FastAPI, Laravel, Yii, Spring Boot, ASP.NET Core, Blazor, Streamlit, Bootstrap, jQuery, Android SDK

AI & Machine Learning: OpenAI GPT-4o, Google Gemini, Vercel AI SDK, LangChain, scikit-learn, Pandas, RAG, Prompt Engineering

Databases: PostgreSQL, MySQL, SQLite, Supabase, phpMyAdmin

DevOps & Cloud: Docker, Docker Compose, Vercel, VPS Deployment, Cloudflare, Azure DevOps, Git, GitHub, GitLab, Bitbucket

Tools & Platforms: n8n, Power BI, RESTful APIs, WSL, Linux, Windows, Visual Studio Code, Android Studio, Jest, React Testing Library

Methodologies: Agile, Scrum, SDLC, CI/CD, Test-Driven Development, Code Review

Certifications: PCEP-30-02 Certified Entry-Level Python Programmer; Microsoft Certified: Azure Fundamentals

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Enerlytic Intelligence Sdn Bhd

Dec 2025 – Present

One City, Subang Jaya

- Developed and deployed a full-stack AI chatbot using Next.js 16, TypeScript, and OpenAI GPT-4o-mini to assess organizational compliance readiness for Malaysia's Energy Efficiency and Conservation Act (EECA) 2024.
- Engineered a structured 10-question assessment flow with automated scoring (0–100), readiness band classification, gap analysis, and lead capture through advanced prompt engineering.
- Built interactive UI components with real-time AI streaming, dynamic option buttons, and multi-select checkboxes using Vercel AI SDK v3 and React 19.
- Integrated PostgreSQL for session tracking, conversation logging, contact management, and user feedback storage.
- Embedded full regulatory text (P.U.(A) 466) into the system prompt, enabling legally accurate responses to EECA compliance queries.
- Containerized the application with multi-stage Docker builds and deployed via Docker Compose with PostgreSQL, supporting self-hosted (VPS + Subdomain) environments.

- Designed a provider-agnostic AI architecture supporting OpenAI, Groq, and Google Gemini with hot-swappable configuration.
- Achieved 54 passing automated tests with Jest and React Testing Library across all components and API routes.

Systems Engineer

Sep 2021 – Nov 2025

Fujitsu Systems Global Solutions Management Sdn Bhd

Menara TM, Kuala Lumpur

- Participated across the full software development lifecycle, including application design, implementation, testing, and troubleshooting.
- Coordinated with global stakeholders, including the Japan Delivery Unit (JDU), to ensure seamless collaboration and solution alignment across time zones.
- Authored PT Specifications listing all test cases to be performed after PG completion, ensuring thorough coverage and traceability.
- Executed manual testing based on PT Specifications and produced PT Evidence to validate functionality and defect-free delivery.
- Performed review activities using PT Review documents, contributing to higher-quality deliverables and a reduced defect rate.
- Developed Proof of Concept (POC) applications and introduced new tools to improve system capabilities and team productivity.

Software Engineer

Apr 2018 – Aug 2021

B2BE GSS Sdn Bhd

Phileo Damansara, Petaling Jaya

- Managed EDI services such as Purchase Orders and Invoices, creating a highly efficient automated and integrated business environment by eliminating manual processing.
- Analyze supplier and buyer requirements for accurate data translation and mapping.
- Implemented requirements by mapping data to the B2BE data structure for seamless cross-system integration.
- Handled mapping validation, ensured documents met trading partner requirements, and supported troubleshooting of failed transactions.
- Tested translations using sample document types and released them to UAT and Production environments.

Software Engineer

Mar 2017 – Mar 2018

Opensoft Technologies Sdn Bhd

Bandar Kinrara, Puchong

- Collaborated with clients to gather documentation and analyze user requirements, producing detailed User Requirements Specification (URS) and Software Requirements Specification (SRS) documents.
- Coordinated User Acceptance Testing (UAT) and Final Acceptance Testing (FAT) to validate functionality and compliance with client expectations.
- Conducted system maintenance, troubleshooting, and debugging to ensure high performance and minimal downtime.
- Applied Agile methodologies and version control (Git) to streamline development and ensure effective team collaboration.

Management Trainee – Network Service Engineer

Aug 2016 – Feb 2017

Telekom Malaysia

Menara TM, Kuala Lumpur

- Troubleshoot various network issues and collaborated with staff and vendors to resolve problems efficiently.
- Configured and installed routers, switches, and wireless controllers across enterprise environments.

- Identified and recommended necessary and optimal infrastructure improvements.

Practical Student – Android Developer

Feb 2016 – Jul 2016

App Lab Sdn Bhd

PFCC, Puchong

- Developed Android applications using Java and the Android SDK.
- Integrated Android applications with back-end services through RESTful APIs.
- Designed and prototyped proof-of-concept solutions to showcase innovative ideas.

KEY PROJECTS

EECA Compliance Assessment Tool

2026

Technologies: Next.js, TypeScript, Linux, VPS, Vercel AI SDK, PostgreSQL, Docker

Developed an AI-powered customer support and compliance assessment chatbot with real-time streaming responses, smart intent classification, lead capture, feedback system, and analytics dashboard. Containerized with Docker, deployed to VPS, and validated with automated test cases.

Smart Energy Anomaly Detector

2026

Technologies: Python, scikit-learn, Docker, Streamlit

Intelligent energy monitoring system using machine learning to detect and explain anomalous power consumption patterns in household energy data.

PulseMetrics — SaaS Analytics Dashboard

2026

Technologies: React 18, Vite, JavaScript, HTML5, CSS3, Recharts

Production-quality analytics dashboard for a fictional subscription business featuring a sleek dark-mode design with glassmorphism effects, gradient accents, and smooth micro-animations.

WhatsApp Food Ordering AI Agent

2025

Technologies: n8n, Docker, Cloudflare, WSL, WAHA (WhatsApp HTTP API), Google Sheets

AI-powered food ordering automation system that enables customers to place orders via WhatsApp, where an AI agent handles conversations, collects order details, and triggers internal restaurant processes.

WBS Whiz — Automated Work Breakdown Structure Generator

2025

Technologies: Python, Pandas, Openpyxl, Tkinter, OpenAI API, Gemini API, Azure DevOps, GitLab

AI-powered desktop application that automatically generates Work Breakdown Structure documents from Fujitsu System Specification files. Parses screen layout and detailed specification Excel files, sends them to Gemini AI for complexity analysis and task scheduling, and produces a professional WBS Excel output (.xlsm) with formatted timelines.

RAG Knowledge Base for SME Operations

2025

Technologies: Python, FastAPI, LangChain, Docker

Built a Retrieval-Augmented Generation system enabling SMEs to securely search and chat with thousands of internal PDFs and SOPs.

Itochu Enex MFG Process Division (WBS Expense System)

2024

Technologies: Vue.js, HTML, CSS

Enhanced user-facing features and built reusable components for the Itochu Enex WBS Expense System, including headers, footers, menu bars, dropdown lists, and overall layout structure.

Toyota / Hakuodo Web Application

2024

Technologies: Python, Django, Pandas, OpenAI API, Azure DevOps

Developed and maintained a web application using Python and Django; debugged and resolved technical issues to ensure system stability.

MFG Electronic — Automated Register Mapping Suite**2022 – 2023****Technologies:** Python, Pandas, Openpyxl, python-pptx, opencv-python, scikit-image (SSIM), Tkinter

Built three automation tools:

- (1) Developed an automated solution by extracting from the register described in the FW request specification definitions file (Excel and PowerPoint) and Unified Register Map (URM) file and mapping them to actual code definitions in ramdef.h file, preventing incorrect register selection and human error;
- (2) Automated register rename updates using ramdef.h, accelerating the Register Map Retrieval process and reducing manual effort when handling large volumes of register changes.; and
- (3) Developed an automated requirement specification diff tool that detects document differences and outputs results to PowerPoint, reducing manual review time and improving implementation and testing efficiency.

Fukuoka Financial Group — Internal Investment Banking System**2022****Technologies:** Java Servlets, JSP, AJAX, Vue.js

Built a web-based internal-use system for a Japanese multinational investment bank.

e-Commerce solutions automating back-office and supply chain**2018 – 2021****Technologies:** C++, XML, EDIFACT, PHP, Git, Bazaar

Managed EDI services such as Purchase Order and Invoice that creates a highly efficient automated and integrated business environment by eliminating manual processing.

E-Vetpermit System — Jabatan Perkhidmatan Veterinar**2017****Technologies:** PHP Yii 2, MySQL, AdminLTE, Bootstrap, jQuery, JavaScript, Git

Permit management system for the transfer of animals between states and within domestic territories for the Malaysian Department of Veterinary Services.

Good Morning Global Android Application**2016****Technologies:** Java, Android SDK, SQLite, Restful Web API, Git

Develop native Android e-commerce application designed to provide users with a seamless, intuitive, and secure mobile shopping experience.

In Case of Emergency (ICE) Android Application**2015****Technologies:** Java, Android SDK, SQLite, Google Map API

Activate an emergency SOS send message (SMS) with a Google Maps link of current location to contacts during an emergency.

EDUCATION

Bachelor of Computer Science (Hons.), Netcentric Computing**2017***Universiti Teknologi MARA (UiTM), Jasin, Melaka***Diploma in Computer Science****2014***Universiti Teknologi MARA (UiTM), Dungun, Terengganu***Sijil Pelajaran Malaysia (SPM) — Mechanical Engineering****2010***Sekolah Menengah Teknik Jasin (SMTJ)*

REFERENCES

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